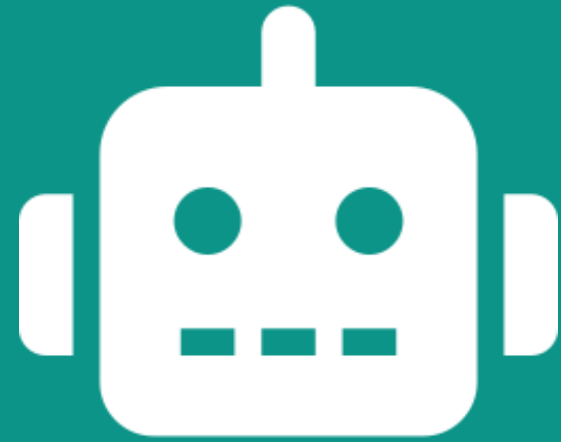




TEAM HAHAIHI

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SmartDesk AI

Confidence-Based Intelligent Ticket Automation
with Human-in-the-Loop

Enterprise AI · Explainable Decisions · Governance-Ready

Presentation Agenda

01

Problem Statement

The scale of enterprise IT ticket pain

02

Proposed Solution

SmartDesk AI — hybrid intelligent system

03

Core Innovation

Confidence engine & explainability

04

Technical Architecture

Pipeline, stack, and governance design

05

MVP Features & Demo

Live system walkthrough

06

Business Impact

ROI, SLA gains, and scalability

07

Limitations & Roadmap

Honest scope + future enhancements

Enterprises are drowning in support tickets - and manual processing is not sustainable.



60-70%

of tickets are
repetitive

Known issues handled manually



SLA

violations
escalating

Response time targets routinely
missed



3×

skilled engineer
time wasted

Experts resolving L1 issues daily



Risk

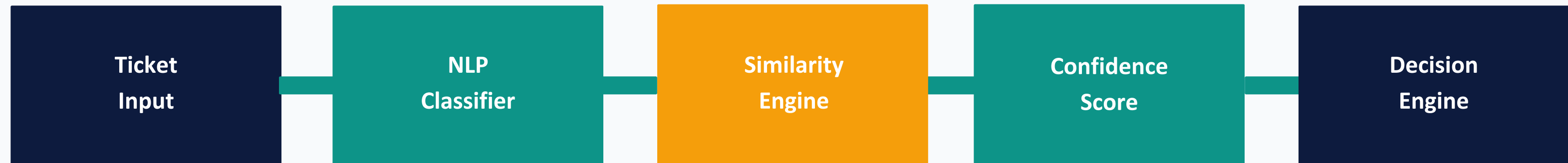
full automation
is unsafe

Critical tickets need human
judgment

Fully automated systems are unsafe for critical or ambiguous issues - enterprises need a smarter, calibrated approach.

SmartDesk AI — How It Works

A hybrid system that automates with intelligence and defers with transparency.



Auto-Resolve

Confidence $\geq 80\%$
System resolves automatically with matched resolution from knowledge base.

Human Review

Confidence $< 80\%$ or Critical
Ticket escalated to agent with AI-suggested resolution and explanation.

P1 / Critical

Always
Human

Confidence Formula: `confidence = (0.6 × similarity_score) + (0.4 × classification_probability)`

Human overrides feed back into the system — continuous learning improves accuracy over time.

Three Pillars of Trust-Calibrated AI



Confidence-Based Decision Engine

- › Dynamic threshold (80%) — not binary on/off (calibrated at 55% for mvp)
- › Weighted score: 60% similarity + 40% classification
- › Priority-aware: P1 always escalates to human
- › Prevents over-automation on ambiguous tickets



Explainable AI Layer

- › Every decision shows similar past tickets
- › Confidence score surfaced to agent in UI
- › TF-IDF feature signals visible per resolution
- › Audit-ready decision log for compliance

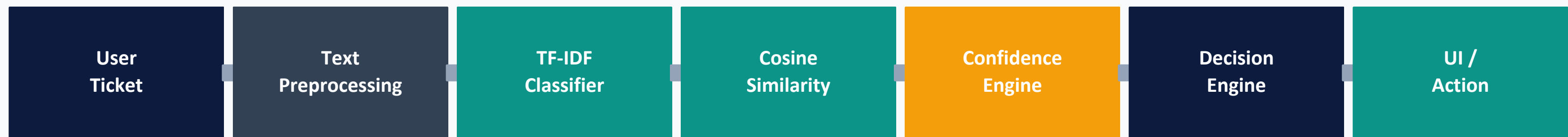


Human-in-the-Loop Learning

- › Agents approve or reject AI suggestions
- › Override decisions captured and stored
- › Model retrain pipeline on accumulated feedback
- › Self-improving support environment

Unlike binary automation, SmartDesk AI calibrates trust dynamically — only acting when confidence warrants it.

System Architecture & Technical Design



Tech Stack & Implementation

Frontend:	Streamlit — single-page interactive UI
NLP:	TF-IDF Vectorizer (Scikit-learn)
Classification:	Logistic Regression (multi-class)
Similarity:	Cosine Similarity
Confidence:	Weighted formula: 0.6 sim + 0.4 prob
Threshold:	0.80 auto-resolve; critical keywords override
Storage:	CSV-based (MVP)
Explainability:	Top-K similar tickets

Governance Controls

- ✓ Configurable confidence threshold (default 80%)
- ✓ Role-based access: User vs Support Agent views
- ✓ Decision log with timestamp, ticket ID, outcome
- ✓ Rollback mechanism — agent can override AI
- ✓ Critical keyword blocklist (outage, production, security breach...)
- ✓ Pending Review Queue — real-time agent dashboard

MVP Features — What's Built & Working



Ticket Submission

User submits ticket via Streamlit form. Category, priority, and text captured.



AI Analysis Engine

NLP classifier + cosine similarity runs on every submission in real time.



Confidence Scoring

Weighted formula produces a score with High / Medium / Low label and color indicator.



Auto-Resolve Flow

Tickets above threshold resolved instantly with matched resolution displayed to user.



Role-Based System

Separate User and Support Agent views with gated access to review queue.



Human Review Queue

Agents see pending tickets with AI suggestion, confidence score, and similar past cases.



Approve/Reject Workflow

Agents approve or override resolution. Decision recorded with timestamp and outcome.



Audit Log + Analytics

Full decision history. Analytics dashboard shows resolution distribution and confidence trends.

Three real scenarios — same system, three different outcomes.

A

Auto-Resolve Case

"Cannot access company VPN after password reset"

Confidence: 87%

AUTO-RESOLVED

Similar ticket found. Resolution: Reset VPN credentials and flush DNS cache. Displayed to user instantly.

B

Human Review Case

"Application throws 500 error intermittently on prod server"

Confidence: 62%

ESCALATED

Below threshold. Agent receives AI suggestion with 3 similar past tickets and confidence breakdown.

C

Critical Override Case

"Security breach detected — unauthorised DB access"

Keyword: security breach

ALWAYS HUMAN

Critical keyword triggers immediate P1 escalation. No confidence check. Sent directly to agent queue.

After demo: Support agent opens Review Queue → approves/rejects → Audit log auto-updates with outcome.

Enterprise Impact & ROI

40–60%

Reduction in
repetitive ticket workload

2–4×

Faster mean time
to resolution (MTTR)

~30%

Cost reduction
in L1 support operations

Operational Efficiency

SLA compliance improvement · Faster L1 resolution · Reduced escalation rate

Cost Optimization

Lower manual triage cost · Expert time freed for L3 · Less reliance on knowledge silos

Governance & Compliance

Explainable AI decisions · Full audit log trail · Configurable risk controls

Strategic Advantage

Scalable managed services · Differentiated AI offering · Faster onboarding for new clients

Why SmartDesk AI — Competitive Landscape

Capability	Traditional Rule Engine	Full AI Automation	SmartDesk AI
Handles repetitive tickets	Partial	Yes	Yes
Confidence-based gating	No	No	Yes - core feature
Critical ticket safety	Manual	Risky	Always human review
Explainable decisions	No	Rarely	Built-in (XAI layer)
Audit log / compliance ready	No	No	Full decision log
Learns from human feedback	No	Batch only	Active learning loop
Priority-aware routing	Static	Implicit	P1/P2/P3 dynamic

SmartDesk AI is the solution that combines confidence-gated automation, explainability, and governance in a single MVP.

MVP Limitations — Acknowledged Constraints

Current MVP Constraints

Training Data:

MVP uses a small synthetic dataset. Real-world accuracy requires enterprise historical data at scale.

Similarity Method:

Cosine similarity on TF-IDF is effective for MVP but lacks semantic depth. SBERT embeddings planned for v2.

Threshold Tuning:

80% confidence threshold is manually calibrated. Production requires data-driven threshold optimization per ticket category.

No Live Integration:

MVP runs standalone (Streamlit + CSV). ServiceNow / Jira integration is a future milestone, not current scope.

Retraining Cycle:

Model retraining is manual in MVP. Active learning pipeline automation is in the roadmap.

What This Demonstrates

- ✓ Strong architectural foundation designed for scale
- ✓ Modular code — each component independently replaceable
- ✓ Governance and auditability built-in from day one
- ✓ Clear upgrade path to production-grade deployment
- ✓ Honest engineering — we know what we built and why



Thank You

SmartDesk AI — Confidence-Based Intelligent Ticket Automation

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